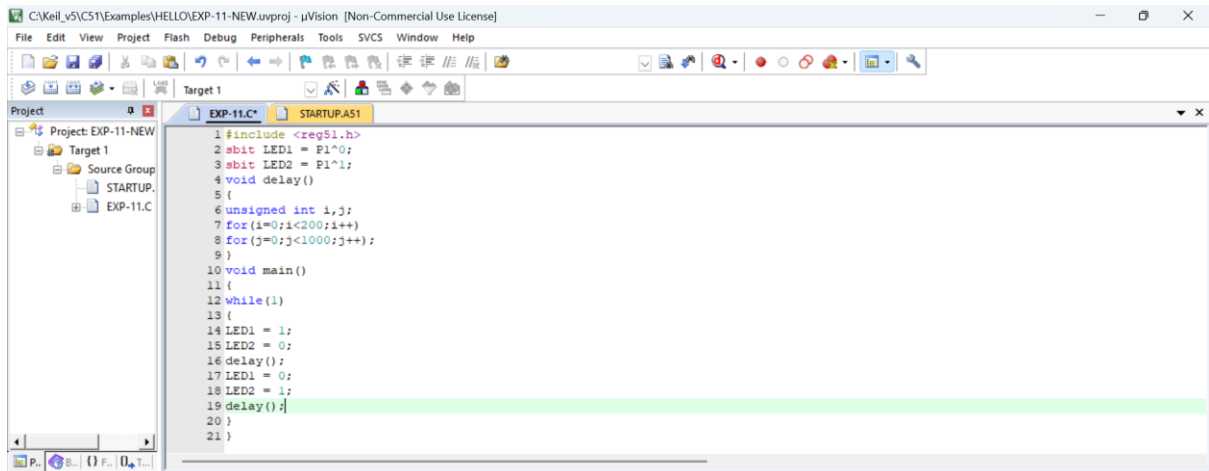
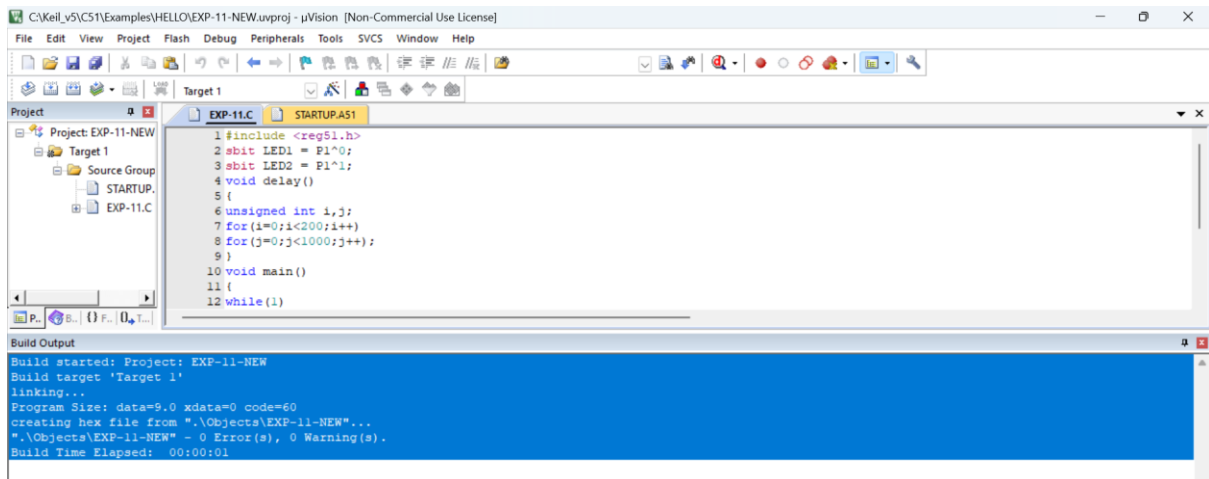


Exp No: 11

STUDY OF GPIO & BIT ADDRESSING IN AT89C51



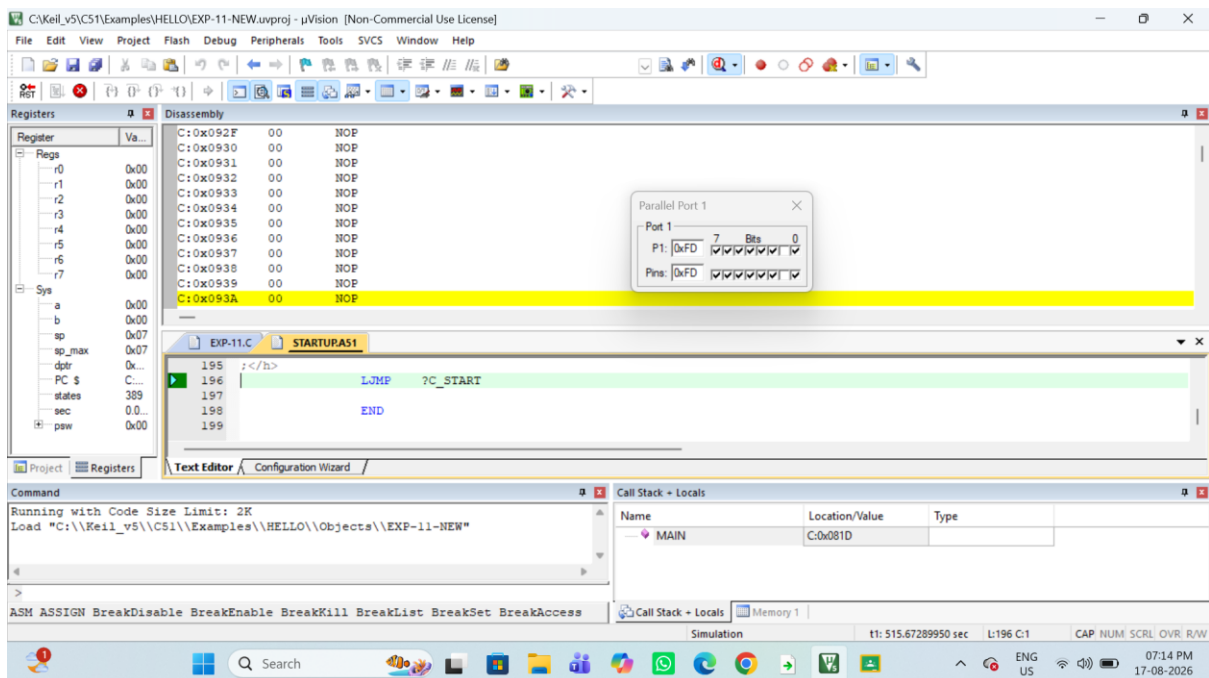
```
1 #include <reg51.h>
2 sbit LED1 = P1^0;
3 sbit LED2 = P1^1;
4 void delay()
5 {
6     unsigned int i,j;
7     for(i=0;i<200;i++);
8     for(j=0;j<1000;j++);
9 }
10 void main()
11 {
12     while(1)
13     {
14         LED1 = 1;
15         LED2 = 0;
16         delay();
17         LED1 = 0;
18         LED2 = 1;
19         delay();
20     }
21 }
```



```
1 #include <reg51.h>
2 sbit LED1 = P1^0;
3 sbit LED2 = P1^1;
4 void delay()
5 {
6     unsigned int i,j;
7     for(i=0;i<200;i++);
8     for(j=0;j<1000;j++);
9 }
10 void main()
11 {
12     while(1)
```

Build Output

```
Build started: Project: EXP-11-NEW
Build target 'Target 1'
linking...
Program Size: data=9.0 xdata=0 code=60
creating hex file from ".\Objects\EXP-11-NEW"...
".\Objects\EXP-11-NEW" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:01
```



Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	0x00
a	0x00
b	0x00
sp	0x07
sp_max	0x07
dptr	0x...
PC	0x...
status	389
sec	0.0...
it	0x00

Disassembly

Address	OpCode	Comment
C:0x092F	00	NOP
C:0x0930	00	NOP
C:0x0931	00	NOP
C:0x0932	00	NOP
C:0x0933	00	NOP
C:0x0934	00	NOP
C:0x0935	00	NOP
C:0x0936	00	NOP
C:0x0937	00	NOP
C:0x0938	00	NOP
C:0x0939	00	NOP
C:0x093A	00	NOP

Parallel Port 1

Port	7	Bits	0
P1:	0xFD	☒	☒
Pins:	0xFD	☒	☒

EXP-11.C

Address	OpCode	Comment
195	:</h>	
196	LJMP	?C_START
197		
198	END	
199		

Command

```
Running with Code Size Limit: 2K
Load "C:\Keil_v5\C51\Examples\HELLO\Objects\EXP-11-NEW"
```

Call Stack - Locals

Name	Location/Value	Type
MAIN	C:0x081D	

Simulation

tt: 515.67289950 sec L:196 C:1 CAP NUM SCRL OVR R/W

07:14 PM 17-08-2026